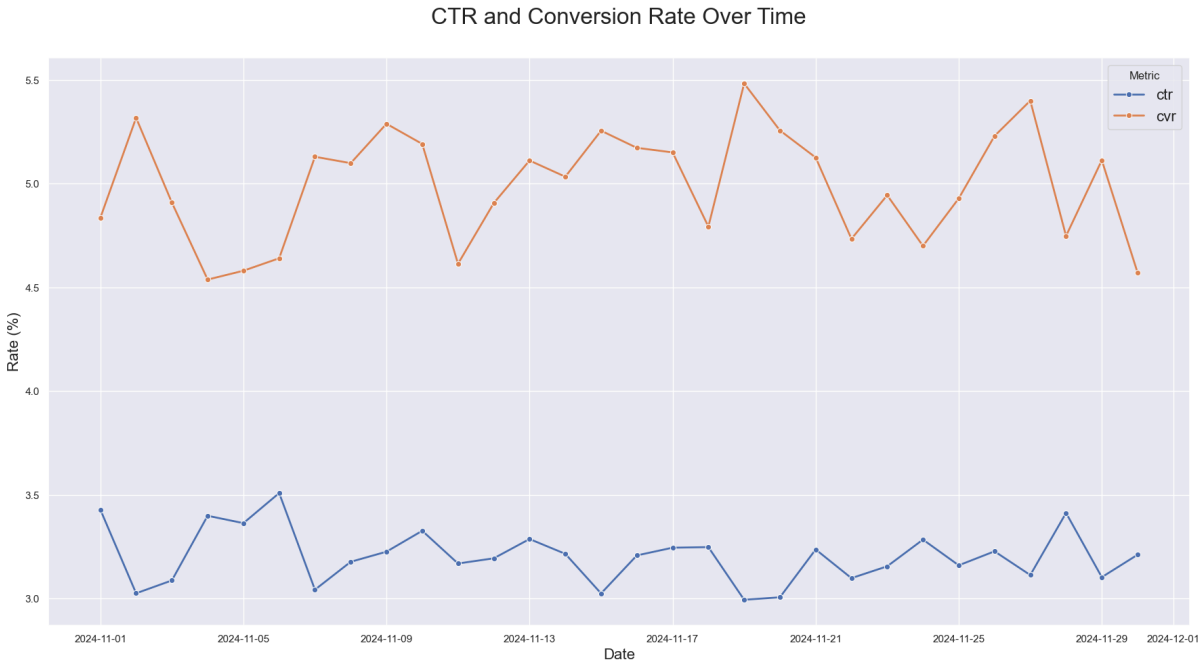
Google Ads A/B Test Design: Generic VS Outcome-Driven Message

Objective

Identify whether changes in Google Ads messaging can improve the stagnated click-through-rate (CTR)

Problem

While the conversion rate (CVR), cost-per-acquisition (CPA), and Return on Ad Spend (ROAS) indicate that the campaign is already efficient (ROAS = 7.03, CPA = 32.96, Revenue-per-click = $10.76), the CTR appears stagnant in November 2024 (~3.1 - 3.4%). Aside from the typos in campaign names, the campaigns were also not very varied, with the keywords, periods, and costs being almost identical.



Assumption

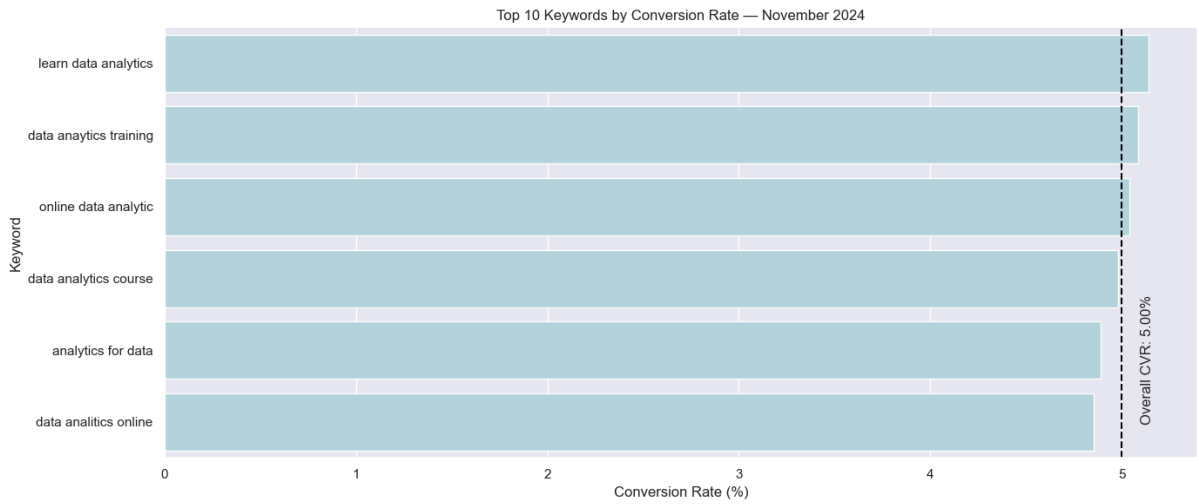
The campaign's headlines and descriptions were generic:

- “Learn data analytics online”
- “Join our data analytics course today”

Customers are more likely to click when the copy is outcome-driven.

Insights

Keywords above the overal CVR: “learn data analytics”, “data analytics training”, and “online data analytic”.



Research Questions
Which ad variation improves CTR without degrading traffic quality?

Hypothesis
H0: There is no significant difference between the current ad copy and the outcome-driven copy.
H1: The outcome-driven copy will generate higher CTR than the current generic ones.

Independent Variable	Dependent Variable
Headline + Description messaging angle	Primary KPI CTR Secondary KPIs - CVR - CPA - ROAS - Revenue per click

Independent Variable	
Variant A (Control)	**Variant B (Treatment)**
- "Learn data analytics online" - "Join our data analytics course today"	"Learn data analytics with real experts" "Data analytics training with certificate" …

Winning Criteria
- +10% relative CTR uplift compared with the control (at least 3.38%)

- CVR should not decrease (>=5%)
- CPA should not increase (<32.96)
- ROAS should not decrease (7.03)

Period
4 weeks